

SHETH LUJ And SIR MV COLLEGE

Practical No 14 and 15

Aim: Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month).

Code:

```
import java.time.LocalDate
import scala.util.Random

case class NiftyData(date: LocalDate, value: Double)

object S091_Shreyash_M2_P14 {

  def main(args: Array[String]): Unit = {

    println("Shreyash Kadam S091")

    val startDate = LocalDate.of(2026, 9, 1)
    val random = new Random(42)

    var currentValue = 25000.0

    val niftyData = (0 until 30).map { day =>

      val date = startDate.plusDays(day.toLong)

      val dailyChange = random.nextDouble() * 400 - 200

      currentValue = currentValue + dailyChange

      NiftyData(date, currentValue)

    }.toList

    println("\n30-Day NIFTY 50 Values:")

    niftyData.foreach { data =>
      println(f"${data.date} : ${data.value}%.2f")
    }

    val averageValue =
      niftyData.map(_.value).sum / niftyData.length

    val highestValue =
      niftyData.maxBy(_.value)
```

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```

val lowestValue =
    niftyData.minBy(_.value)

println("\nTime Series Analysis:")

println(f"Average NIFTY Value: $averageValue%.2f")

println(
    f"Highest NIFTY Value: ${highestValue.value}%.2f on ${highestValue.date}"
)

println(
    f"Lowest NIFTY Value: ${lowestValue.value}%.2f on ${lowestValue.date}"
)

println("\n7-Day Moving Average:")

val movingAverage =
    niftyData.sliding(7).map { window =>

        val avg =
            window.map(_.value).sum / window.size

        (window.last.date, avg)

    }.toList

movingAverage.foreach { case (date, avg) =>
    println(f"$date : $avg%.2f")
}

val firstValue = niftyData.head.value
val lastValue = niftyData.last.value

println("\nTrend Analysis:")

if (lastValue > firstValue)
    println("NIFTY shows an increasing trend.")
else if (lastValue < firstValue)
    println("NIFTY shows a decreasing trend.")
else
    println("NIFTY remains almost stable.")
}
}

```

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Output:

```
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
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30-Day NIFTY 50 Values:
2026-09-01 : 25091.03
2026-09-02 : 25164.31
2026-09-03 : 25087.80
2026-09-04 : 24998.63
2026-09-05 : 25064.85
2026-09-06 : 25226.20
2026-09-07 : 25173.72
2026-09-08 : 25084.01
2026-09-09 : 25069.48
2026-09-10 : 25182.64
2026-09-11 : 25350.37
2026-09-12 : 25324.96
2026-09-13 : 25424.93
2026-09-14 : 25379.55
2026-09-15 : 25250.51
2026-09-16 : 25288.25
2026-09-17 : 25172.15
2026-09-18 : 25302.54
2026-09-19 : 25171.43
2026-09-20 : 25206.40
2026-09-21 : 25306.91
2026-09-22 : 25335.32
2026-09-23 : 25367.33
2026-09-24 : 25468.34
2026-09-25 : 25280.91
2026-09-26 : 25224.07
2026-09-27 : 25351.19
2026-09-28 : 25318.27
2026-09-29 : 25507.88
2026-09-30 : 25593.24

Time Series Analysis:
Average NIFTY Value: 25258.91
Highest NIFTY Value: 25593.24 on 2026-09-30
Lowest NIFTY Value: 24998.63 on 2026-09-04

7-Day Moving Average:
2026-09-07 : 25115.22
2026-09-08 : 25114.22
2026-09-09 : 25100.67
2026-09-10 : 25114.22
2026-09-11 : 25164.47
```

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```
Time Series Analysis:  
Average NIFTY Value: 25258.91  
Highest NIFTY Value: 25593.24 on 2026-09-30  
Lowest NIFTY Value: 24998.63 on 2026-09-04
```

```
7-Day Moving Average:
```

```
2026-09-07 : 25115.22  
2026-09-08 : 25114.22  
2026-09-09 : 25100.67  
2026-09-10 : 25114.22  
2026-09-11 : 25164.47  
2026-09-12 : 25201.63  
2026-09-13 : 25230.01  
2026-09-14 : 25259.42  
2026-09-15 : 25283.20  
2026-09-16 : 25314.46  
2026-09-17 : 25312.96  
2026-09-18 : 25306.13  
2026-09-19 : 25284.19  
2026-09-20 : 25252.97  
2026-09-21 : 25242.60  
2026-09-22 : 25254.71  
2026-09-23 : 25266.01  
2026-09-24 : 25308.32  
2026-09-25 : 25305.23  
2026-09-26 : 25312.75  
2026-09-27 : 25333.44  
2026-09-28 : 25335.06  
2026-09-29 : 25359.71  
2026-09-30 : 25391.99
```

```
Trend Analysis:
```

```
NIFTY shows an increasing trend.
```

```
Process finished with exit code 0
```

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Aim: Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3 (e.g., [1, 1², 1³, 2, 2², 2³, 3, 3², 3³]).

Code:

```
object S091_Shreyash_PolynomialFeatures {

  def main(args: Array[String]): Unit = {

    println("Shreyash Kadam S091")

    val data = List(1, 2, 3, 4, 5)

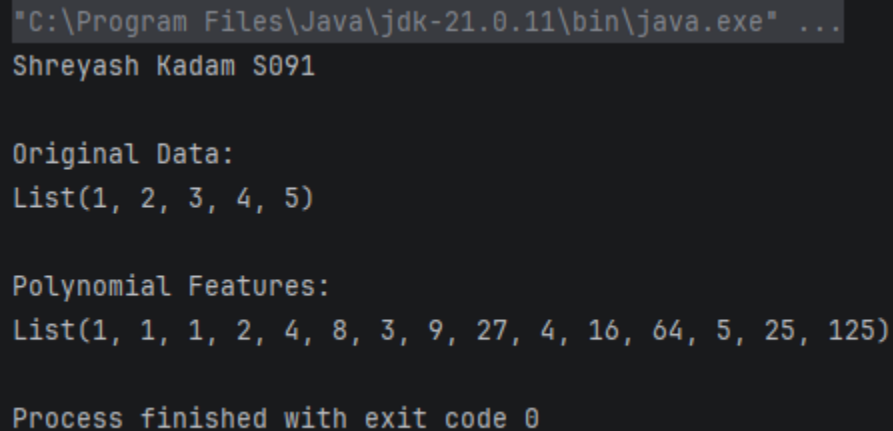
    val degree = 3

    val polynomialFeatures = data.flatMap { x =>
      (1 to degree).map(d => math.pow(x, d).toInt)
    }

    println("\nOriginal Data:")
    println(data)

    println("\nPolynomial Features:")
    println(polynomialFeatures)
  }
}
```

Output:



```
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
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Original Data:
List(1, 2, 3, 4, 5)

Polynomial Features:
List(1, 1, 1, 2, 4, 8, 3, 9, 27, 4, 16, 64, 5, 25, 125)

Process finished with exit code 0
```